

remotely. Such IoT switches can also be wirelessly connected to an IoT hub or router.

- **Voltage sensors:** IoT voltage sensors can monitor the supply voltage of the house remotely, using the Internet.
- **Energy monitors:** IoT energy monitors are digital energy meters that can monitor the power consumption of the house, and this can be done remotely.
- **Thermostats:** IoT thermostats can be used to monitor the temperature of the house in real-time, and via the Internet; this can be done remotely.
- **Smart door locks:** These can be controlled using a home security system and IoT OSS or apps, over the Internet.
- **Air conditioners:** Controlling the air conditioner remotely is a key part of home automation. IoT based ACs can be automated and controlled remotely using an IoT OSS or app connected to the Internet.
- **Surveillance cameras:** IoT cameras can be used for the surveillance of the house and one can view live video footage of the house remotely. Both indoor and outdoor cameras can be used to monitor both environments. Such a surveillance system can have features like motion detection and can alert home owners whenever a security breach occurs. Many IoT surveillance OSS and apps can be used to enhance the security and monitoring system of the house. Many IoT based home appliances like washing machines, water heaters, refrigerators, dish washers, robot floor cleaners, etc, can also be used to improve the automation of the house.

Controller/hub

The controller/hub used for the IoT devices is the main part of the home automation system because it controls all the IoT devices/sensors installed in a house. The connection can be either wired or wireless though the

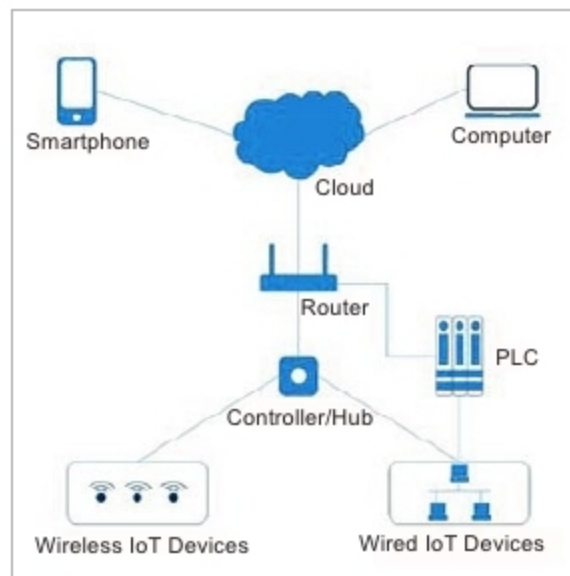


Figure 1: Home automation setup

latter option is better. The controller is connected to the router (gateway) of the house for Internet connectivity. The IoT controller/hub uses mainly three types of standards to communicate with IoT devices – ZigBee, Z-Wave and Wi-Fi. Wink and SmartThings are the two most common types of IoT controllers used for home automation. With such a controller/hub, it's possible to connect hundreds of IoT devices/sensors in a house, which in

turn can be controlled with the help of a smartphone app, automatically.

FOSS for home automation

There are various free and open source software (FOSS) available for home automation. The most popular ones are featured here.

- **OpenHAB:** This FOSS for home automation is written in Java. It is used to connect and control various IoT devices installed in a house. With this app, users can perform many operations like switching on lights and fans remotely. They can apply various rules to program (automate) the IoT devices and can also use voice commands to control them.

The Eclipse Foundation began developing this project in 2010. In 2013, the core functionality became an official project of the foundation and was called Eclipse SmartHome. OpenHAB is not only available for Android OS but also for other systems like Linux, Windows,



Figure 2: IoT devices